

**Uncertainty, Innovation, and Migration: A Macro-Level Empirical Investigation of
Inventor Mobility Under Uncertainty**

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Abstract

Inventors represent a distinctive category of high-skilled knowledge workers whose migration decisions are shaped by drivers different from those influencing the broader category of skilled labour. This study examines how macro-level uncertainty affects cross-border inventor mobility, using a panel of bilateral flows between 122 origin and 122 destination countries from 1993 to 2012. Employing a gravity-type framework, the analysis focuses on relative uncertainty between destination and origin countries, rather than on absolute levels in isolation. The estimates show that a one-standard-deviation increase in the relative log uncertainty (destination over origin) is associated with a 19.7% to 25.2% rise in inventor outflows. Split-sample results indicate that this effect is concentrated in cases where uncertainty in the origin country is markedly higher than in the destination, underscoring the importance of pronounced stability gaps as a migration driver. The relationship remains robust to the inclusion of innovation-specific controls and the specifications in which macroeconomic scale effects are absorbed by high-dimensional fixed effects, as well as to an alternative uncertainty measure. These findings challenge the central prediction of Real Options Theory—that uncertainty delays irreversible decisions—by revealing a behavioural mechanism in which inventors exit unstable environments promptly when more stable alternatives exist abroad. The study extends the literature on high-skilled migration and the geography of innovation under uncertainty, offering new evidence that relative, directional asymmetries in macro-level risk can act as a decisive push trigger for globally mobile inventors.

Keywords: Gravity-type model, Innovation systems, Inventor migration, Push migration, Real-Options Theory, Relative uncertainty

1 Introduction

The international migration of inventors plays a central role in shaping the global geography of innovation and cross-border diffusion of knowledge. Unlike general skilled migration, inventor mobility entails not only the relocation of human capital but also the transfer of tacit knowledge, research practices, and technology-specific capabilities. These flows can alter regional comparative advantage, deepen transnational knowledge networks, and influence long-term trajectories of scientific productivity. Understanding who migrates, and under what conditions, is therefore of central importance to policymakers, firms, and scholars concerned with the future of innovation systems.

A growing body of empirical research has established that mobile inventors tend to be more productive, more internationally collaborative, and more embedded in global scientific networks than their non-mobile peers. Prior studies attribute such mobility to a range of structural and institutional pull factors—most notably, high research and development (R&D) expenditure—that enhance the attractiveness of destination countries. Yet, the role of macro-level uncertainty remains largely unexplored.

Prevailing studies often treat uncertainty as an exogenous risk factor, implicitly assuming it impacts migration by discouraging investment and reducing the innovative activity in the research ecosystem (Ivus & Wajda, 2018). This logic aligns with classical Real-Options Theory, which posits that under uncertainty, rational agents delay irreversible decisions until greater information clarity is achieved (Bloom, 2009). However, applying this framework to high-skilled and globally mobile inventors may be overly restrictive. Their internationally portable human capital, embedded in transnational research networks, enables movement across institutions, sectors and geographies with comparatively low sunk cost. As a result, inventor migration is only partially irreversible, reducing the incentive to postpone relocation in the face of uncertainty. What remains unresolved is the behavioural mechanism

driving this response: whether heightened origin-country instability primarily pushes inventors out, or whether the relative stability of alternative destinations exerts a stronger pull.

Building on this premise, the present study hypothesises that inventor outmigration is shaped less by the absolute level of uncertainty and more by its relative distribution across countries. Specifically it posits that a higher macro-level uncertainty in the origin country may trigger outward movement, especially when the destination country is comparatively stable. Rather than treating uncertainty as a uniform deterrent, this analysis investigates how divergence in uncertainty—its relative distribution—shapes the geography of high-skilled migration.

To evaluate this hypothesis, the study uses global panel of bilateral inventor migration flows between 122 origin and 122 destination countries from 1993 to 2012. The key explanatory variable is a ratio-based measure of relative uncertainty, which captures directional asymmetries in perceived risk between sending and receiving countries. A gravity-type estimation framework with three-way fixed effects and lagged bilateral innovation-specific controls are employed, with macroeconomic scale effects (GDP per capita and population) absorbed by the fixed effects, to isolate the effect of uncertainty from other core drivers of inventor mobility.

Rather than treating uncertainty as a static implicit factor in migration decisions, this study examines it as an active driver of inventor outflows. Prior research, such as Bisset et al. (2024), has explored uncertainty as a push factor, but their analysis is limited to a discrete-choice framework using individual-level inventor data across 12 European countries. In contrast, this study adopts a macro-structural perspective, analysing how asymmetries in national uncertainty levels influence global inventor flows between 122 origin and destination countries.

The empirical findings presented here show that relative uncertainty matters: inventor outflows intensify when origin-country risk rises in the presence of substantially greater stability elsewhere, underscoring the role of asymmetric macro-volatility in shaping global talent flows. These results point not only to the behavioural distinctiveness of inventors but also to the need for more targeted talent attraction strategies in an increasingly volatile global environment.

2 Literature Review

The study of inventor migration has traditionally been situated within the broader literature on high-skilled mobility, yet it presents distinctive characteristics. Inventors differ from general knowledge workers in that they produce frontier technologies, and their international movements have direct implications for the spatial distribution of innovation capacity and the cross-border diffusion of tacit knowledge. Thus, inventor migration warrants a separate analytical treatment.

Pioneering work by Miguelez and Fink (2013, 2017) introduced a global, patent-based methodology for tracking inventor migration through changes in PCT filing addresses. Their data revealed a sustained rise in cross-border inventor mobility since the mid-1980s, with flows that follow patterned, path-dependent trajectories shaped by colonial ties, geographic proximity, and diasporic networks.

Akbaritabar et al. (2024) significantly expands this approach by documenting bilateral inventor migration across 210 countries from 1998 to 2020 using Scopus data. Their findings indicate that mobility is increasingly concentrated in innovation-intensive economies, and that migration patterns often mirror scientific collaboration networks and the international distribution of research output. The consequences of such mobility have been the subject of growing empirical attention.

Using an event-study design, Pellegrino et al. (2023) demonstrate that inventor migration is associated with substantial productivity gains. On average, inventors generate 0.23 more patent applications in the five years following relocation, with larger gains observed when moving to destinations with stronger intellectual property regimes and more developed research infrastructure.

Migueluez and Temgoua (2020) further show that knowledge flows are not unidirectional: origin countries also benefit, particularly through return migration, co-invention, and diasporic linkages. These findings support the view that inventor migration facilitates international knowledge recombination and technological spillovers, rather than simply reallocating talent from Global South to Global North.

However, scientific opportunity structures alone cannot fully explain inventor migration. Recent work emphasises the role of institutional frictions—particularly policy uncertainty and legal restrictions—in shaping mobility decisions. Soto Nishimura and Czaika (2024), in a meta-analysis of high-skilled migration studies, report consistent negative elasticities between migration policy restrictiveness and skilled inflows.

Similarly, Kahn and MacGarvie (2020), in their analysis of STEM PhDs trained in the United States, find that delays in permanent residency processing significantly reduce retention. For Chinese nationals, each additional year of delay reduces the probability of staying by 2.1 percentage points; for Indian nationals facing wait times exceeding five years, the likelihood of staying declines by over 10 percentage points. These findings underscore the critical role of institutional openness in the retention high-skilled migrants.

Despite these advances, the role of uncertainty—economic, political, or institutional—remains comparatively underexamined in the inventor migration literature. Classical models rooted in Real-Options Theory suggest that agents delay irreversible decisions under uncertainty to preserve future flexibility (Dixit & Pindyck, 1994; Bloom, 2009).

Empirical evidence supports this logic: firms typically reduce or postpone R&D investment when faced with heightened economic policy uncertainty (Czarnitzki & Toole, 2011; Ivus & Wajda, 2018). Such models implicitly assume that uncertainty suppresses forward-looking action, functioning as a brake on both innovation and cross-border mobility of resources.

However, this logic may not hold for individuals – particularly inventors – who possess internationally portable human capital and face relatively low sunk costs of relocation. Unlike firms, inventors operate across jurisdictions, collaborate within distributed global networks, and respond to a distinct set of constraints and incentives. Their behavioural responses to uncertainty may therefore diverge from firm-level decision-making.

Bisset et al. (2024) offer the first migration-specific test of this hypothesis. Using a binary choice model across twelve European countries, they find that a one-standard deviation (SD) increase in economic policy uncertainty at the origin raises the probability of inventor outmigration between 13.73% and 21.70%. Interpreted as push effect, these findings suggest that uncertainty can accelerate skilled emigration, particularly from institutional environments perceived as volatile or deteriorating.

Nevertheless, important limitations remain. The scope of analysis in Bisset et al. (2024) is confined to twelve European countries, where harmonised mobility regimes under the Schengen framework minimise institutional barriers to migration. Moreover, their study relies on Economic Policy Uncertainty (EPU) index, which captures policy-related volatility but omits broader macroeconomic and geopolitical risks addressed by the World Uncertainty Index (WUI). Their empirical strategy also estimates the probability of leaving the origin country using inventor-level patent data, without modelling destination choice. Additionally, their measure of uncertainty is constructed by comparing each origin country's uncertainty level to a time-specific cross-country average, rather than to the uncertainty of specific

destination countries. This framing captures general push effects but does not model directional movement or bilateral comparisons. As such, it remains unclear whether uncertainty operates solely as a repelling force, or whether inventors systematically compare uncertainty across countries before moving. Nor does this framework reveal whether destination's own low uncertainty encourages inflows or is neutralised by even higher risk at the origin.

This study addresses these gaps by extending the empirical frame to 122 origin and 122 destination countries, and by constructing a relative log uncertainty ratio using the WUI (Ahir et al., 2022). This ratio-based specification evaluates how bilateral asymmetries in macro-level volatility shape inventor migration—asking not simply whether inventors leave unstable countries, but whether they do so only when a meaningfully more stable option exists abroad. To isolate this effect, the empirical strategy incorporates innovation-specific controls and implements a three-way fixed effects structure that also absorbs macroeconomic scale effects and other unobserved characteristics at both country and dyad levels.

By doing so, this study advances a more nuanced understanding of how macro-level uncertainty shapes the global mobility of high-skilled human capital. It challenges the universality of Real Options Theory and introduces relative risk perception as a critical determinant of cross-border talent flows—especially salient in a post-crisis world marked by institutional volatility and intensifying competition for innovation.

3 Methodology

Gravity-type frameworks provide the work-horse structure for analysing bilateral migration flows of skilled workers. Following Beine et al. (2016) and Fink et al. (2017), this study models inventor movements as a function of relative push-pull forces between 122 origin and 122 destination countries. The estimation strategy is aligned with best-practice guidelines for

high-dimensional gravity estimation, as set out by Yotov et al. (2016), and accounts for a rich set of bilateral and country-specific heterogeneities.

The dependent variable, $Flow_{ijw}$, denotes the cumulative inventor flow from origin country i to destination country j during five-year window w . It is defined as:

$$Flow_{ijw} = \sum_{t=n}^{n+4} Flow_{ijt}$$

where $n \in \{1993, 1998, 2003, 2008\}$ and $Flow_{ijt}$ is the count of inventors who, in year t , list country i as their country of origin and country j as their country of residence on a PCT patent application. This data comes from inventor mobility dataset compiled by Miguelez and Fink (2013, 2017). Aggregating over non-overlapping five-year windows reduces noise in annual fluctuations and enables a stable comparison of bilateral migration patterns over time, in contrast to the use of five-year moving averages.

The key explanatory variable is a measure of relative log-uncertainty between destination and origin countries, calculated as:

$$rWUI_{ij(w-1)} = \frac{\ln WUI_{j(w-1)}}{\ln WUI_{i(w-1)}}$$

where $WUI_{j(w-1)}$ and $WUI_{i(w-1)}$ denote the uncertainty values for the destination and the origin countries, respectively, averaged over the five-year window preceding w . The explanatory variable is lagged by one window ($w - 1$), to mitigate reverse causality bias—that is, to ensure that the conditions influencing migration are temporally prior to the observed inventor flow (Fink et al., 2017). Uncertainty and innovation conditions measured in $w - 1$ can plausibly shape the inventor decisions realised in window w , but migration flows in w cannot feed back into already-lagged regressors.

The relationship between inventor flows and relative uncertainty is modelled using the following multiplicative gravity-type specification:

$$Flow_{ijw} = \exp (\alpha_{ij} + \alpha_{iw} + \alpha_{jw} + \beta rWUI_{ij(w-1)} + \delta rX_{ij(w-1)} + \epsilon_{ijw}), \quad (1)$$

where, α_{ij} captures time-invariant bilateral factors such as geographic distance, common language, and colonial ties; α_{iw} represents origin-window fixed effects, which capture time-varying conditions in origin countries—such as domestic policy shifts, economic growth, or political instability that may influence emigration rates; and α_{jw} denotes destination-window fixed effects, which accounts for time-varying conditions in destination countries, including changes in immigration policy, research funding, or institutional reform. Together, these fixed effects absorb unobserved heterogeneity at both the bilateral and country-window level, allowing β to be interpreted as the directional effect of relative uncertainty within dyads over time. A positive coefficient on $rWUI$ implies that inventor outflows increase as the uncertainty asymmetry between origin and destination increases.

The specification also includes several innovation-specific controls, all incorporated in relative log ratios, $rX_{ij(w-1)}$, mirroring the transformation applied to the key uncertainty regressor. Macroeconomic scale variables such as GDP per capita and population are not included as separate regressors; instead, their relative log ratios are fully absorbed by the three-way fixed effects. This ensures that cross-country differences in economic size or market potential are accounted for implicitly, while estimated coefficients reflect directional push-pull contrasts between country pairs rather than absolute levels.

The equation (1) is estimated using Poisson Pseudo Maximum Likelihood (PPML), the preferred estimator for gravity-type models involving heteroscedastic count data and structural zero flows (Head & Mayer, 2014). The inclusion of three-way fixed effects absorbs time-invariant bilateral characteristics as well as time-varying origin- and destination-specific unobservable characteristics, ensuring that identification is achieved through variation in relative uncertainty and innovation conditions within dyads over time. Standard errors are clustered at the origin-destination level to allow for arbitrary serial correlation within dyads.

To facilitate the interpretation, the coefficient on $rWUI$ is standardised such that a one-unit increase corresponds to a one-SD change in the sample, following Bisset et al. (2024). Accordingly, coefficient β represents the percentage change in inventor flows associated with a one-SD increase in relative log uncertainty, conditional on bilateral innovation differences and fixed effects.

To test the validity of the results, two robustness checks are also conducted. First, the model is re-estimated using Ordinary Least Square (OLS). While less appropriate for skewed, zero-inflated data, OLS provides a direct estimator comparison with PPML using the same three-way fixed effects, with macroeconomic variation absorbed in the fixed effects rather than entered as separate controls. Second, the regressors are smoothed using five-year moving averages to assess the persistence of uncertainty effects beyond short-term volatility. These robustness exercises are also presented in the Results section.

4 Data and Variable Definitions

Inventor flows are sourced from bilateral migration data compiled by Miguelez and Fink (2013, 2017), which records, for each origin-destination pair, the number of inventors who, at the time of a PCT patent application in year t , self-report country j as their residence and country i as their country of origin.

These bilateral counts are aggregated into four five-year windows (1993-1997, 1998-2002, 2003-2007, and 2008-2012), resulting in a panel dataset of 59,048 dyad-window observations covering 122 origin and 122 destination economies. An initial inspection of the raw data reveals several stylised facts that motivates the empirical strategy adopted in this study.

Table 1 presents the ten largest destination countries for migrant inventors across the four windows. The United States dominates consistently, absorbing over 85,000 inventors during the 2008-2012 period – more than eight times its intake during 1993-1997. Germany,

Switzerland, and the United Kingdom also rank highly throughout, although their relative shares decline over time. Notably, China enters the top ten only in the final window, indicating an emergent role as a destination in the global innovation landscape

Table 2 presents the complementary perspective, listing the leading origin countries. While the United Kingdom and Germany dominated in the early periods, China and India exhibit significant growth, jointly accounting for a combined 48,000 outbound inventors by 2008-2012. These shifts reflect a broader reconfiguration of the global innovation system, with talent being increasingly sourced from emerging economies.

To explore how these national trends manifest at the bilateral level, Table 3 reports the ten busiest inventor migration corridors in the earliest and latest windows. The China-United States corridor expands more than tenfold—from just over 2,000 to nearly 23,000 migrants—emerging as the dominant global inventor mobility channel by the end of the sample period. Other major corridors, such as India-United States, Canada-United States, and Germany-Switzerland also grow, although their relative share of total flows narrows. These patterns underscore the importance of modelling both persistent bilateral linkages and evolving push-pull dynamics through a gravity-type framework.

Consistent with the modelling strategy, all explanatory and control variables are expressed as relative log ratios. For each variable, the value for the destination country is divided by the corresponding value for the origin country, and both terms are transformed using the natural logarithm prior to forming the ratio. This specification aligns the scale of all regressors with key uncertainty measure and facilitates interpretation in terms of directional push-pull contrasts between country pairs.

The principal explanatory variable, $rWUI$, is derived from the World Uncertainty Index (Ahir et al., 2022), which quantifies macroeconomic, political, and institutional

Table 1 *Top Ten Destination Countries of Migrant Inventors during Four Five-Year Time Windows*

Country	Inflows	Country	Inflows	Country	Inflows	Country	Inflows
	1993-1997		1998-2002		2003-2007		2008-2012
US	10,346	US	36,354	US	97,829	US	85,579
Germany	2,636	Germany	6,364	Germany	11,713	Germany	11,377
UK	1,873	UK	4,972	Switzerland	9,360	Switzerland	10,714
Switzerland	1,833	Switzerland	4,792	UK	7,507	UK	7,096
France	1,211	France	2,591	Netherlands	4,772	France	4,188
Australia	867	Netherlands	2,369	France	4,617	China	4,177
Canada	739	Canada	1,955	Canada	3,471	Singapore	3,618
Belgium	674	Australia	1,605	Singapore	3,098	Netherlands	3,441
Sweden	510	Japan	1,452	Japan	2,726	Japan	2,930
Japan	435	Belgium	1,433	Belgium	2,337	Sweden	2,828

Table 2 *Top Ten Origin Countries of Migrant Inventors during Four Five-Year Time Windows*

Country	Outflows	Country	Outflows	Country	Outflows	Country	Outflows
	1993-1997		1998-2002		2003-2007		2008-2012
UK	3,790	China	10,618	China	27,616	China	27,874
China	2,712	UK	8,281	India	21,602	India	20,470
Germany	2,549	Germany	8,180	Germany	16,691	Germany	14,816
India	1,823	India	6,536	UK	15,187	UK	11,607
US	1,258	France	4,416	Canada	11,325	Canada	10,543
Canada	1,116	Canada	4,229	France	9,087	France	10,122
France	1,057	US	3,012	US	5,397	US	5,912
Austria	883	Italy	2,307	Netherlands	4,966	Italy	5,604
Netherlands	831	Netherlands	2,159	Italy	4,517	Netherlands	3,856
Italy	722	Russia	2,026	Russia	4,247	Japan	3,310

Table 3 *Top Ten Bilateral Inventor Migration Corridors during Two Five-Year Time Window*

Origin	Destination	Flows	Origin	Destination	Flows
		1993-1997			2008-2012
China	US	2,039	China	US	22,885
UK	US	1,744	India	US	17,621
India	US	1,572	Canada	US	8,969
Canada	US	884	UK	US	6,334
Germany	Switzerland	697	Germany	US	4,624
Germany	US	691	Germany	Switzerland	4,177
Austria	Germany	619	France	US	3,102
UK	Germany	328	Japan	US	2,274
UK	Australia	289	France	Switzerland	1,872
Russia	US	280	Russia	US	1,573

uncertainty based on the frequency of uncertainty-related terms in EIU country reports. For each origin-destination pair, it is ensured that the uncertainty conditions measured are temporally prior to the inventor flows they are intended to explain.

In addition, the model includes a set of control variables capturing structural features of national innovation systems: international collaboration, total citation counts, scholarly output and number of authors (as proxy for inventor stock). These data are sourced from SciVal, Scopus, and Akbaritabar et al. (2024). Each control variable is constructed analogously to the key regressor, as a relative log ratio averaged over the same lagged five-year window.

Table 4 provides summary statistics for the key explanatory variables. The distribution of inventor flows is highly right skewed, with a mean of 7.03 and a maximum

Table 4 *Descriptive Statistics*

Variable	Observations	Mean	SD	CoV*	Min	Max
Bilateral Inventor Migration Flow	59,048	7.03	203.93	29.02	0	23523
Relative Log Uncertainty	54,932	1.09	0.48	0.44	0.18	5.53
Relative Log Citation	44,286	1.12	0.81	0.73	0.04	23.20
Relative Log Int. Collaboration	44,165	1.21	1.16	0.95	-15.81	23.60
Relative Log Output	44,286	1.28	1.28	1	0.03	31.60
Relative Log Num of Inventors	43,566	1.21	0.87	0.72	0.07	15.27
Relative Log GDP Per Capita PPP	58,564	1	0.03	0.03	0.91	1.09
Relative Log Population	59,048	1.01	0.12	0.12	0.62	1.60

CoV* = Coefficient of Variation

Table 5 *Correlation Matrix*

	1	2	3	4	5	6	7	8
1	1.000							
2	-0.009	1.000						
3	0.004	-0.129	1.000					
4	-0.003	-0.116	0.926	1.000				
5	-0.003	-0.100	0.904	0.911	1.000			
6	-0.003	-0.111	0.900	0.918	0.990	1.000		
7	0.007	0.007	-0.112	-0.104	-0.100	-0.105	1.000	
8	-0.001	-0.160	0.405	0.331	0.394	0.412	-0.094	1.000

Note. 1 = bilateral inventor flows; 2 = relative log uncertainty; 3 = relative log citations; 4 = relative log collaborations; 5 = relative log output; 6 = relative log num of inventors; 7 = relative log gdp per capita ppp; 8 = relative log population

exceeding 23,000, justifying the use of PPML over OLS for estimation. Relative uncertainty near or above 1. These patterns validate the inclusion of innovation-specific metrics in the gravity model to capture meaningful bilateral differences.

Finally, Table 5 presents the correlation matrix for the explanatory variables. As expected, the innovation-specific controls – including relative citations, collaboration, output, and inventor stock – exhibit strong pairwise correlations, frequently exceeding 0.90. Importantly, however, relative uncertainty is only weakly correlated with the control variables. This orthogonality suggests that the main coefficient of interest – β on $rWUI$ – is not confounded by underlying scientific fundamentals or unobserved institutional quality.

5 Results

5.1 Relative Uncertainty as a Determinant

With the empirical framework in place, the analysis first examines the role of relative uncertainty as a determinant of inventor migration. The coefficient on the key regressor, $rWUI$, indicates whether inventor flows respond systematically to differences in macro-level volatility between destination and origin countries.

Table 6 presents a stepwise estimation strategy, gradually introducing fixed effects to control for unobserved heterogeneity. Column (1) presents a naïve bivariate regression of inventor flows on relative log uncertainty ratio, with no fixed effects. The coefficient on $rWUI$ is negative and weakly significant, suggesting that, on average, inventor migration maybe lower when relative log uncertainty is higher. However, this specification does not account for unobserved heterogeneity and is likely biased due to omitted variables correlated with both uncertainty and mobility—such as global crisis periods or systemic shocks.

Column (2) introduces origin-destination fixed effects, absorbing time-invariant country-pair specific characteristics. In this specification, the coefficient on $rWUI$ is identified using over-time variation in the relative uncertainty between origin and destination

countries. Moreover, because no country-window fixed effects are included at this stage, the estimate also captures time-varying conditions at both the origin and destination, common to all world partners—such as shifts in domestic innovation policy or broader macroeconomic trends—that are correlated with uncertainty. The negative and statistically significant coefficient thus reflects both movements in relative uncertainty and any co-evolving national-level dynamics not yet accounted for in the model.

Column (3) also adds origin-window fixed effects alongside dyadic fixed effects. These absorb time-varying conditions in origin-countries that are common to all partner destination countries. The coefficient on $rWUI$ is identified from within-dyad changes in the relative log uncertainty ratio across time, conditional on stable origin conditions. The estimate turns positive but remains statistically insignificant, suggesting no clear association between shifts in relative uncertainty and inventor flows under this specification.

Column (4) instead includes destination-window fixed effects along with the dyadic fixed effects, controlling for time-varying shocks at the destination, that are common to all partner origin countries. The effect of $rWUI$ is now identified from changes in the relative log uncertainty ratio where destination conditions are held constant. The coefficient becomes negative and statistically significant, indicating that inventor flows decrease when the relative uncertainty ratio rises. While the direction reverses compared to Column (3), this shift reflects sensitivity to country-specific volatility and underscores the need to control for both sides of the dyad simultaneously.

Column (5) includes the full set of three-way fixed effects: dyadic, origin-window, and destination-window. The saturated specification absorbs time-invariant bilateral characteristics and time-varying origin-specific and destination-specific shocks, such that identification arises solely from within-dyad variation in the relative log uncertainty ratio over time. The coefficient on $rWUI$ is positive and statistically significant at the 10% level.

Table 6 *PPML Estimation with Introduction of Fixed Effects*

VARIABLES	(1) No FE	(2) Single FE	(3) Two-way FE	(4) Two-way FE	(5) Three-way FE
Relative Log Uncertainty	-0.531*	-0.795***	0.128	-0.133*	0.410*
	(0.278)	(0.286)	(0.110)	(0.069)	(0.210)
Constant	2.572***	7.638***	6.902***	7.164***	6.636***
	(0.442)	(0.282)	(0.108)	(0.0691)	(0.208)
Observations	54,932	8,873	8,738	8,651	8,517
Origin-Destination FE	No	Yes	Yes	Yes	Yes
Origin-Time FE	No	No	Yes	No	Yes
Destination-Time FE	No	No	No	Yes	Yes
Pseudo-R ²	0.005	0.928	0.982	0.983	0.986

Note. Columns correspond to alternative fixed-effects specifications. (1) All cross-sectional and time-series variation without absorbing fixed effects; (2) variation over time within each origin-destination pair, across destinations for a given origin, and across origins for a given destination; (3) variation in origin over time, differing across destination partners; (4) variation in destination over time, differing across origin partners; (5) variations within each destination-origin pair over-time.

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This indicates that holding fixed bilateral frictions and country-specific dynamics, inventor flows increase in periods where the relative uncertainty between destination and origin countries becomes more pronounced.

However, as the regressor is specified as a log-ratio, the coefficient in Column (5) captures relative asymmetry rather than directional change. The observed association may be driven by rising volatility at the destination, declining uncertainty at the origin, or combination of both. What the estimate reflects is a response to growing disparity in macro-level uncertainty between countries, even after controlling for structural and temporal factors.

To probe whether this effect is primarily push- or pull-driven, Table 7 disaggregates the sample based on directional asymmetry in country-level uncertainty. Column (1) restricts the sample to dyads where the origin exhibits above-average uncertainty and the destination is below average; Column (2) captures the reverse configuration. The results show a strong, positive, statistically significant association between the relative log uncertainty ratio and inventor flows only in Column (1), where uncertainty is elevated at the origin. In contrast, when the destination is more uncertain than the destination (Column 2), the coefficient is small and insignificant.

Column (3) and (4) present two additional reference groups: dyads where both origin and destination are highly uncertain, and dyads where both are stable. In both cases, the coefficient on $rWUI$ is statistically insignificant and negative, suggesting that shared levels of risk—whether high or low—do not generate a systematic migratory response. These results reinforce the interpretation that inventor flows are responsive not to absolute uncertainty per se, but to directional asymmetries—particularly those marked by heightened origin uncertainty relative to stable destinations. This confirms that the key mechanism is relative push effect, rather than a symmetric risk aversion.

Such findings help clarify how the magnitude of the estimated coefficient in Table 6 Column (5), should be interpreted. The coefficient of 0.410 indicates that a one-unit increase in the relative log uncertainty ratio between destination and origin countries is associated with a 41% increase in inventor flows, conditional on fixed dyadic and country-level factors. Following the interpretation strategy proposed by Bisset et al. (2024), scaling the coefficient by the within sample standard deviation of $rWUI$ ($SD = 0.48$) yields an implied elasticity of approximately 19.7%. This suggests that a one-SD increase in relative log uncertainty ratio is associated with a nearly 20% increase in inventor flows. In light of the sample-split results, however, this effect appears to reflect heightened inventor mobility in response to worsening

Table 7 *Sample-Split Estimation by Directional Asymmetry in Uncertainty*

	(1)	(2)	(3)	(4)
VARIABLES	High Ori Low Des	Low Ori High Des	High Ori High Des	Low Ori Low Des
Relative Log Uncertainty	1.416** (0.618)	0.172 (2.063)	-0.036 (0.730)	-2.721 (1.736)
Constant	3.949*** (0.876)	8.586*** (1.457)	7.006*** (0.793)	8.810*** (1.705)
Observations	1,167	1,077	1,212	1,135
Three-way FE	Yes	Yes	Yes	Yes
Pseudo-R ²	0.996	0.996	0.979	0.979

Note. Columns correspond to subsamples defined by directional asymmetries in uncertainty, where “high” and “low” classifications are assigned relative to the mean WUI for origin and destination countries with each five-year window. (1) Dyads where origin uncertainty is above and destination uncertainty is below the window-specific mean; (2) dyads where origin uncertainty is below and destination uncertainty is above the window-specific mean; (3) dyads where both origin and destination uncertainty exceed the window-specific mean; (4) dyads where both origin and destination uncertainty fall below the window-specific mean.

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

uncertainty in the origin context—rather than any systematic attraction to volatile destinations.

This result is corroborated and strengthened in the fully saturated specification of Table 8, which retains benchmark three-way fixed effects and includes all innovation-specific controls simultaneously, with macroeconomic scale effects absorbed by the fixed effects. Even in this restrictive model, the coefficient on relative log uncertainty remains positive, statistically significant, and virtually unchanged in magnitude as 0.532. Interpreted at one-SD, this estimate implies a 25.5% increase in inventor outflows from high-uncertainty origin countries. The persistence of this effect in the most stringent model specification reinforces the central finding from Table 6 & 7: inventor migration systematically rises with heightened uncertainty at the origin, consistent with uncertainty acting as a push trigger, not deterrent.

Table 8 *PPML Estimation with Innovation-specific Controls*

VARIABLES	(1) Network Effect	(2) Ecosystem Strength	(3) Potential Movers	(4) Ecosystem Quality	(5) All Controls
Relative Log Uncertainty	0.491** (0.225)	0.507** (0.225)	0.506** (0.225)	0.526** (0.223)	0.532** (0.222)
Relative Log Int. Collaboration	0.354 (0.362)				-1.025 (0.965)
Relative Log Output		1.098** (0.534)			1.774 (1.318)
Relative Log Num of Inventors			1.016 (0.658)		-0.648 (1.235)
Relative Log Citation				2.484 (1.819)	2.597 (3.402)
Constant	6.265*** (0.465)	5.396*** (0.637)	5.508*** (0.767)	3.816* (2.072)	3.570 (3.141)
Observations	6,576	6,578	6,564	6,578	6,564
Three-way FE	Yes	Yes	Yes	Yes	Yes
Pseudo-R ²	0.987	0.987	0.987	0.987	0.987

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

5.2 Impact of Controls

To ensure that the estimated effect of relative uncertainty is not confounded by omitted drivers of inventor migration, the specification augments the baseline with a set of innovation-specific and macroeconomic controls identified in the literature. These include counts of international collaboration, scholarly output, citations, and inventor stock. The objective here is not to advance new theoretical claims regarding these covariates, but to verify that the relative-uncertainty effect persists once these factors are accounted for, and that their signs are consistent with established empirical findings.

Table 8 introduces the innovation-specific controls sequentially along with the three-way fixed effects in all columns. This step-wise approach mitigates the omitted variable bias by conditioning on time-varying unobservables at both origin and destination. Across Columns (1) to (4), the coefficient on relative log uncertainty remains stable, positive, and significant, while each control behaves as theory predicts: greater international collaboration, higher research output, a larger inventor pool, and stronger citation performance at the destination (relative to origin) are all associated with higher inventor flows. These patterns mirror evidence of Fink et al. (2017) and Gorin (2016) that scientific connectivity, ecosystem capacity, and technological reputation are important pull factors.

Column (5) includes all innovation-related controls simultaneously. Although multicollinearity limits the interpretability of individual coefficients in this saturated model, the stability of the relative log uncertainty coefficient confirms that its effect is orthogonal to these ecosystem attributes, highlighting robustness under maximum conditioning. Overall, the PPML estimates in Table 8 confirm that the link between inventor migration and relative uncertainty is not an artefact of unmodelled macroeconomic scale and scientific capacity. Having established that the effect survives strict controls, the analysis next turns to robustness checks that compare estimation frameworks and assess the role of temporal dynamics.

5.3 Sensitivity and Robustness Checks

Two complementary robustness exercises assess the stability of the estimated relationship between relative log uncertainty and inventor migration. First, the analysis compares results across estimation frameworks by re-estimating selected PPML specifications from Tables 6-8 using Ordinary Least Squares (OLS) under the same three-way fixed-effects structure (Table 9). Second, the uncertainty measure is redefined as a five-year moving average to test whether inventor flows respond more to persistent volatility than to discrete shocks (Table 10).

In Table 9, Columns (1) – (5) mirror the baseline and split-sample PPML specifications, and Column (6) mirrors the saturated specification with all scientific-specific controls. Across all six OLS regressions, the sign and approximate magnitude of the relative log uncertainty coefficient are broadly consistent with the corresponding PPML estimates, although OLS produces larger standard errors and, in some cases, wider confidence intervals. This pattern reflects OLS's limitations with sparse, heteroscedastic migration-flow data, yet the qualitative consistency across estimators reinforces that the main relationships are not dependent on the estimation method.

Table 10 probes temporal robustness by replacing the lagged non-overlapping window measure of relative uncertainty with a lagged five-year moving average. This smoother specification tests whether inventor flows respond to longer-run, background volatility rather than short-term uncertainty shocks. Across all variants, the coefficient on the moving-average is small, statistically insignificant, and economically negligible—implying less than a 4.8% change in flows per standard deviation. This contrasts with the more substantial effects observed using the standard lagged measure, suggesting that inventor mobility is primarily triggered by short-term directional changes rather than by chronic, slow-moving risk.

Table 9 *OLS Estimation as Robustness Check*

VARIABLES	(1) Base	(2) High Ori Low Des	(3) Low Ori High Des	(4) High Ori High Des	(5) Low Ori Low Des	(6) All Controls
Relative Log Uncertainty	4.026 (3.240)	0.520 (4.093)	28.71 (21.95)	-9.115 (7.065)	4.637 (12.78)	1.651 (3.106)
Relative Log Int. Collaboration						-46.24*** (12.67)
Relative Log Output						30.72** (14.86)
Relative Log Num of Inventors						-9.276 (13.88)
Relative Log Citation						226.1*** (60.61)
Constant	3.160 (3.534)	4.791 (6.953)	-4.752 (13.87)	19.05*** (7.361)	-0.595 (13.16)	-206.8*** (58.51)
Observations	54,930	9,346	9,346	6,402	13,124	43,080
R-squared	0.728	0.828	0.998	0.993	0.782	0.879
Three-way FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted-R ²	0.619	0.608	0.608	0.608	0.608	0.811

Note. Columns correspond to OLS estimation, mirroring earlier PPML specifications. (1) Baseline model with three-way fixed effects; (2) – (5) split-sample estimations; (6) saturated model including all scientific-specific controls.

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 10 *PPML Estimation with Scientific Controls as Five-year Moving Averages*

VARIABLES	(1) Two-way FE	(2) Two-way FE	(3) Three-way FE	(4) Three-way FE
Relative Log Uncertainty	0.099 (0.079)	-0.268*** (0.068)	0.024 (0.196)	0.023 (0.201)
Relative Log Int. Collaboration				-1.751 (2.141)
Relative Log Output				5.987 (3.796)
Relative Log Num of Inventors				-3.622 (2.363)
Relative Log Citation				-0.769 (4.758)
Constant	5.380*** (0.077)	5.741*** (0.066)	5.472*** (0.189)	5.735* (3.418)
Observations	42,672	41,500	39,029	30,573
Origin_Destination FE	Yes	Yes	Yes	Yes
Origin-Time FE	Yes	No	Yes	Yes
Destination-Time FE	No	Yes	Yes	Yes
Pseudo-R ²	0.957	0.959	0.962	0.965

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Taken together, the results from Table 9 and 10 show that core findings are not an artefact of estimator choice or temporal framing. The alignment between PPML and OLS estimates under identical fixed-effects structures indicates that the positive association between relative uncertainty and inventor migration is qualitatively robust to alternative estimation methods. The moving-average exercise further suggests that inventor mobility is driven primarily by short-term directional changes in uncertainty rather than by chronic, slow-moving volatility. Overall, the evidence supports the interpretation that inventors respond decisively to recent macroeconomic uncertainty shocks at the origin, treating them as push triggers rather than conditions warranting delay – contrary to the conventional predictions of Real Options Theory.

6 Discussion and Conclusion

This study examines how macro-level uncertainty influences the global migration of inventors, focusing on directional asymmetries between origin and destination countries. Across specifications, the analysis finds that inventor flows rise significantly when uncertainty increases in the origin country relative to the destination. A one-SD increase in the relative uncertainty measure is associated with a 19.7% to 25.2% increase in bilateral inventor outflows. This effect is statistically significant, robust to the inclusion of innovation-specific controls and stable across estimation frameworks. However, it is driven overwhelmingly by scenarios in which uncertainty is elevated in the origin country and comparatively low in the destination. When the destination itself is highly uncertain, the effect dissipates or turns negative.

These results suggest that inventors are prompted to move when their origin country becomes more uncertain relative to viable alternatives abroad. Uncertainty operates not as a universal deterrent, but as a directional push mechanism – a force that triggers emigration only when a meaningful risk gradient exists. This interpretation aligns with recent findings by

Bisset et al. (2024), who document higher emigration probabilities among individual European inventors facing elevated economic policy uncertainty. However, the present study extends and reframes their argument by embedding it in a global, bilateral gravity framework and showing that destination-side uncertainty does not exert a corresponding pull.

This finding partially diverges from the predictions of Real Options Theory. Traditional models posit that agents delay irreversible decisions under uncertainty to preserve flexibility (Dixit & Pindyck, 1994; Bloom, 2009). If inventor migration were governed by this logic, one would expect uncertainty to suppress movement. Instead, the results indicate that inventors act decisively in the face of rising volatility – exiting unstable countries without delay. Importantly, however, the effect disappears when uncertainty is smoothed using five-year moving averages. This implies that inventors respond to acute, short-run shocks rather than to persistent background fragility – a temporal logic that remains consistent with Real Options Theory, but a behavioural logic that diverges from it. Inventors do not wait for clarity; they respond to turbulence – yet only when a more stable alternative exists.

Robustness checks further affirm the credibility of this interpretation. Across both PPML and OLS estimators, the coefficient on relative log uncertainty retains its sign and direction while the effect of long-run smoothed uncertainty remains negligible. GDP per capita and population are absorbed by the fixed effects in all specifications and therefore do not alter the main result, indicating that the behavioural response is not simply a function of economic scale or market size. The structural feature that matters most is temporal: inventors react to short-term uncertainty, not to chronic, slow-moving risk.

Nonetheless, several limitations warrant acknowledgement. First, the use of three-way fixed effects, while essential for controlling unobserved heterogeneity, absorbs much of the structural variation in macroeconomic indicators, limiting the capacity to test dynamic policy

effects or model heterogeneous responses. Second, the inventor flow data, while rich, exclude informal migration, non-patenting researchers, and temporary mobility – thereby narrowing the scope of inference. Third, only a small share of dyad-window observations contain both non-zero flows and usable variation in the regressor, reducing statistical power for detecting subgroup effects.

Future work could address these limitations through the use of broader researcher mobility datasets such as Scopus-based author affiliations by Akbaritabar et al. (2024), which capture a wider spectrum of scientific migration. Linking migration to post-move patent productivity, collaboration networks, or innovation outcomes could also reveal whether exiting uncertain environments yield scientific gains. Finally, survey-based or experimental designs could illuminate whether inventors consciously interpret macro-level uncertainty as a threat, an opportunity, or both.

In conclusion, this study shows that macro-level uncertainty, when measured in relative and directional terms, serves as a behavioural push trigger for high-skilled inventors. Inventors do not flee risk indiscriminately, nor are they drawn to it. Instead, they respond when the asymmetry between origin and destination conditions becomes too wide to ignore. In doing so, they convert uncertainty into movement – not by delaying action, but by exiting early.

For inventors, uncertainty is not a theoretical variable to wait out, but a lived condition. Embedded in global networks, equipped with mobile human capital, and attuned to shifts to institutional stability, they are among the first to respond when relative uncertainty rises. They do not wait for domestic investments to rebound or reform signals to materialise. They move – decisively, often permanently. If uncertainty continues to intensify globally and countries fail to offer relative stability, the risk is not merely the loss of innovators, but the

erosion of the future innovation they enable. In this light, uncertainty is not just a macro-level concern; it is a matter of talent retention, scientific continuity, and long-run competitiveness.

APPENDIX

List of Countries

OECD countries (31): Australia, Austria, Belgium, Canada, Chile, Colombia, Costa Rica, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Israel, Italy, Japan, Latvia, Lithuania, Mexico, Netherlands, New Zealand, Norway, Poland, Portugal, Slovenia, Spain, Sweden, Switzerland, United Kingdom, United States.

Non-OECD countries (91): Afghanistan, Albania, Algeria, Angola, Argentina, Armenia, Azerbaijan, Bangladesh, Benin, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Central African Republic, Chad, China, Croatia, Dominican Republic, Ecuador, El Salvador, Eritrea, Ethiopia, Gabon, Georgia, Ghana, Guatemala, Guinea, Guinea-Bissau, Honduras, India, Indonesia, Iraq, Jamaica, Jordan, Kazakhstan, Kenya, Kuwait, Lebanon, Lesotho, Liberia, Libya, Madagascar, Malawi, Malaysia, Mali, Mauritania, Moldova, Mongolia, Morocco, Mozambique, Myanmar, Namibia, Nepal, Nicaragua, Niger, Nigeria, Oman, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Qatar, Russian Federation, Rwanda, Saudi Arabia, Senegal, Singapore, South Africa, Sri Lanka, Sudan, Tajikistan, Tanzania, Thailand, Togo, Tunisia, Turkmenistan, Uganda, Ukraine, United Arab Emirates, Uruguay, Uzbekistan, Viet Nam, Zambia, Zimbabwe

Notes

1. PCT stands for “Patent Cooperation Treaty”

2. EIU stands for “Economist Intelligence Unit”

$$3. Flow_{ijs}^{MA} = \frac{1}{5} * \sum_{s=t-4}^t Flow_{ijs}, \text{ where } t \in [1993, 2012]$$

$$4. \overline{WUI}_{cw} = \frac{1}{N_w^c} * \sum_{c \in C_w} (WUI_{cw}), \text{ where } c \in \{i, j\}$$

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